

$A_{120} - \theta = 2/9$: Location, Status, and Limit

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Contents

.1	The Question	1
.2	The Magnitude Channel Is Blind	1
.3	The Elimination Chain	1
.4	Forced or Contingent	2
.5	The Geometric Source: C_3 in A_5	2
.6	Verification Script	3
.7	Symbol Glossary	3
.8	Open Questions	3
.9	Supersedes	4
.10	Use of AI Tools	4

.1 The Question

$\theta = 2/9$ is empirically confirmed to seven figures (A110). This document shows three things: *where* the value resides (solely in the dynamic phase channel, not in the magnitude), *that* it follows exactly and parameter-free from the icosahedral doubling geometry, and *what* remains open as a derivation step (the bridge to the mass ratios).

The central finding is positive: two mutually independent paths – the phenomenological Koide angle from the measured masses and the purely geometric redistribution of modes – hit exactly the same rational number $2/9$, without knowing about each other.

.2 The Magnitude Channel Is Blind

[K] With amplitudes $a_k(\theta) = 1 + \sqrt{2} \cos(\theta + 2\pi k/3)$ the Koide invariant is

$$Q(\theta) = \frac{\sum_k a_k^2}{(\sum_k a_k)^2} = \frac{6}{9} = \frac{2}{3} \quad \text{for every } \theta.$$

The proof is two lines: three cosines offset by 120° sum to zero, hence $\sum_k a_k = 3$; and with $\sum_k \cos^2 = 3/2$ it follows that $\sum_k a_k^2 = 6$.

[K] From this follows a statement to be recorded, because it limits the reach of A110: *the Koide relation says nothing about θ* . $Q = 2/3$ holds for every angle. The value $2/9 = 0.2222$ also lies *below* the range $[1/3, 1]$ that Q traverses for positive masses – $2/9$ is therefore neither a limit, nor an extremum, nor a fixed point of the symmetric functional.

In the magnitude channel $2/9$ is completely invisible.

.3 The Elimination Chain

[K] Systematically checked where θ does *not* live. The orbit members $\{1/9, 2/9, 4/9\}$ are to be distinguished; the question is which quantity achieves this.

Channel	Behaviour	Distinguishes?
Magnitude $ B_\theta $	$= 1$ to 10^{-15} for all θ	no
Winding number	$= 3$ exactly for all θ	no
Symmetric functional	identical for all θ	no
Continuous phase profile	0.093 / 0.171 / 0.249	yes

[K] Noteworthy is the second row. The integer winding – precisely the topological quantity that, according to A030, carries the information concept of the theory – does *not separate* $2/9$ from $1/9$ and $4/9$. The reason is that $2/(9\pi)$ is irrational; θ is not a flat topological invariant.

[K] Exactly one channel remains: the continuous, dynamic phase profile. There and only there is $2/9$ distinguishable from its orbit companions.

.4 Forced or Contingent

[K] This raises the decisive question: is $2/9$ forced like the speed of light, or contingent?

The test is the variational behaviour. Coupling a second triplet with strength δ , the extremum moves away from the Z_3 point: θ^* rises smoothly and monotonically from 0.099 at $\delta = 0.10$ through 0.223 at $\delta = 0.24$ to 0.501 at $\delta = 1.0$. The achieved range $[0.05; 0.52]$ contains $2/9$ *in the interior and undistinguished*.

[K] This test sets δ as a *free* coupling parameter. Under this assumption $2/9$ would lie undistinguished in the achieved range and the value would be contingent. But the assumption is not that of the theory: the icosahedral geometry fixes δ (next section). Once δ is no longer free but the total leak $(7 - 3\varphi)/9$ of the same redistribution that also carries θ , the contingency falls away. The variation test therefore shows that θ *lives in the dynamic channel* – it does not decide forced/contingent, because it assumes δ to be free, which the geometry does not.

.5 The Geometric Source: C_3 in A_5

[B] The dynamic channel has a geometric source. The three lepton modes are the Fourier eigenmodes of the Z_3 circulant: the trivial mode $v_0 = \frac{1}{\sqrt{3}}(1, 1, 1)$ and two non-trivial, Galois-conjugate modes. The doubling places the C_3 structure into the icosahedral group A_5 ; its fivefold generator is a rotation R_5 by 72° about an icosahedral fivefold axis. Because the threefold and fivefold axes are differently oriented, R_5 mixes the C_3 modes.

[B] The complete redistribution of the electron mode has weights $p_j = |\langle v_j | R_5 v_e \rangle|^2$:

$$\underbrace{p_0 = \frac{2}{9}}_{\text{trivial}}, \quad p_2 = \frac{5-3\varphi}{9}, \quad p_1 = \frac{2+3\varphi}{9}, \quad p_0 + p_1 + p_2 = 1,$$

with $\varphi = \frac{1+\sqrt{5}}{2}$. All three weights carry the denominator $9 = 3^2$ – the same Z_3 structure in which $2/9$ sits as orbit address. The fraction that transitions into the trivial mode under the fivefold rotation is

$$p_0 = |\langle v_0 | R_5 v_e \rangle|^2 = \frac{2}{9}$$

exactly and parameter-free (numerically confirmed to 40 places). No mass ratio, no Koide parametrisation, and no $2/9$ input enters this value.

[K] The value is icosahedron-specific, not generic:

- 200 random rotation axes at 72° give the leak in the range $[0.036; 0.452]$; *not one* hits $2/9$. Only the icosahedral fivefold axes deliver it.
- The angle scan shows $2/9$ exactly at 72° ; the n -fold series gives $2/9$ exclusively at $n = 5$ (at $n = 3$ one gets $2/5$, at 144° one gets $4/9$).
- Of the six fivefold axes, three give $2/9$, three give $4/9$; the embedding selects the nearest axis and hence $2/9$. That is a discrete binary choice, not a free parameter.

[B] The same redistribution also delivers the second quantity: the total leak $p_0 + p_2 = (7-3\varphi)/9 = 0.2384$ is the parameter-free value of the coupling δ and agrees with the $\delta^* = 0.2389$ back-computed from θ to 0.18 %. Angle *and* amplitude therefore arise from a single fivefold redistribution; the earlier separate treatment as two harmonics was too narrow a view. With the tolerance from the local slope (0.052 % instead of 0.135 %), the leak $(7-3\varphi)/9$ lies about 3.5σ from δ^* and is not compatible with it; the total leak is therefore a candidate for δ , not its value. $\theta = p_0 = 2/9$ is unaffected.

[K] Thus two independent paths coincide exactly: the phenomenological Koide angle (from the masses, without geometry) and the geometric redistribution (from A_5 , without masses and without $2/9$ input) both give $2/9$. That is a convergence finding. $2/9$ is an invariant of the translation between torus geometry and spectral representation – it belongs to the common structure, not to one representation, analogously to $\hbar$ and c being conversion factors and not ontological primitives.

[B] The connection to the dynamic channel closes here: the fivefold rotation R_5 sets a phase $\pi/2$ – precisely the value $\chi = \pi/2$ that marks the one distinguishing channel above. The geometric source and the dynamic location are the same.

.6 Verification Script

- θ -independence of Q (exact, for arbitrary angles), the elimination chain through the three blind channels, and the contingency test $\theta^*(\delta)$ with the proof that $2/9$ lies undistinguished in the achieved range: `python/A_Serie_Skripte/a120_theta.py`

.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\theta = 2/9$	Geometric fixed-point angle; dynamic locus on T^4
$p_0 = 2/9$	Exact θ value as C_3 -in- A_5 identity
Ω_{res}	Resonance frequency of the θ -mode
$\Delta\omega$	Frequency spacing to the next mode

.8 Open Questions

First, the value $2/9$ is derived, not open. It follows exactly and parameter-free from the icosahedral redistribution and independently coincides with the phenomenological Koide angle. The earlier classification as structurally unjustified is thereby superseded.

Second, δ is not free. It is the total leak $(7 - 3\varphi)/9$ of the same redistribution that carries θ . The contingency impression arose solely from treating δ as a free parameter; the geometry fixes it. The leak is, however, not compatible with δ^* ($\approx 3.5\sigma$; see the section on the geometric source); which value fixes δ is therefore an open point.

Third, one derivation edge remains open – and is precisely named. The explicit mechanism connecting the geometric redistribution weights $(2/9, (5 - 3\varphi)/9, (2 + 3\varphi)/9)$ with the physical mass ratios has not been written out. This bridge need not run via Koide; it can be led directly through the masses or the mode structure. That is the subject of further work – and explicitly not an objection to the convergence finding, but the next stage. A125 works out this bridge as far as the corpus closes it: one matrix element of R_5 carries the Koide formula, and the φ skeleton of the weights, $p_1/p_2 = \varphi^8$, $p_0/p_2 = 2\varphi^4$, hits the ratio m_τ/m_e to 3×10^{-5} with $(74 + \frac{1}{45})\varphi^8$; the factors 74 and $1/45$ are observed, not derived.

Fourth, the underlying doubling structure A_5 is presupposed. The value $2/9$ is exact within the established icosahedral doubling; whether this doubling itself necessarily follows from the three stipulations of A020 or enters as an additional structure is the remaining structural question – one that goes beyond the five-fold counting but no longer touches the value $2/9$.

.9 Supersedes

This document subsumes: Doc. 291 (dynamic locus), Doc. 293 (icosahedral approach), Doc. 286 (dynamic sector), Doc. 290 (magnitude/phase, insofar as concerning θ). Incorporated: P42, P35, R139.

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.